

Radar HRV Monitoring with Physiological Prior Inspired Deep Neural Networks

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Abstract—Radar sensing has emerged as a promising solution for the contactless monitoring of Heart Rate Variability (HRV), a crucial indicator of the cardiovascular and autonomic nervous systems. However, due to signal noise and interference that easily obscure heartbeat details, along with variations in heartbeat across different physiological conditions, existing methods remain restricted to laboratory settings with healthy subjects and fail in real-world scenarios involving more complex physiological conditions. In this study, we propose a physiological prior-inspired deep learning framework for robust radar-based HRV monitoring. Specifically, we leverage the prior that internal heartbeats drive movements across the entire torso surface and design a hybrid deep neural network to model the spatio-temporal relationship between full-body radio reflections and heartbeats, effectively mitigating interference. Then, we incorporate the cardiac motion's self-similarity prior to establish a signal augmentation strategy, effectively remodeling the HRV distribution and enhancing performance across diverse physiological conditions. We build and validate our method on a large-scale dataset comprising 7,150 outpatients with complex physiological conditions in real-world scenarios. The experimental results demonstrate that our method achieves a mean IBI error of 19.21 ms, an RMSSD error of 16.23 ms, an SDSD error of 16.70 ms, and a pNN50 error of 7.28%. We further validate the performance by classifying five common cardiac conditions based on HRV results, demonstrating performance comparable to ECG-based methods. These results highlight the great potential of our approach for accurate, contactless HRV monitoring in real-world applications.

Index Terms—HRV, Radar Sensing, Deep Neural Networks

I. INTRODUCTION

IN recent years, cardiovascular diseases have become a major global health concern due to their profound impact on individual well-being and public health systems [1]–[3]. Heart Rate Variability (HRV), the variation in intervals between heartbeats, is a vital biomarker for evaluating cardiovascular health, playing an important role in early disease

This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by the Medical Research Institutional Review Board (IRB) of the First Affiliated Hospital of USTC under Application No. 2023KY056.

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prevention and health management [4], [5]. Therefore, there has been growing research interest in developing convenient, comfortable, and effective methods for HRV monitoring [6]–[8]. Among these advancements, radar-based HRV monitoring stands out for its ability to provide contactless, privacy-preserving, and comfortable monitoring, along with cost-effective deployment [9]–[23].

The working principle of radar-based HRV monitoring is straightforward. Internal cardiac motions drive corresponding movements on the body surface. By analyzing these movements, which modulate the phase of radio signals reflected from the human body, detailed HRV information can be obtained. Building on this principle, signal processing-based approaches have demonstrated notable performance [24]–[31]. However, these methods rely heavily on predefined assumptions about signal characteristics, such as fixed frequency bands or limited mode bandwidths. In real-world scenarios, where signal characteristics are highly dynamic, these assumptions often fail, resulting in significant performance degradation. Recently, with the rise of deep learning, researchers have made significant strides by directly mapping radar signals to cardiac waveforms, achieving further improved performance in HRV monitoring [32]–[38]. For instance, Zhang et al. [32] proposed a deep generative model designed to decompose nonlinear signal mixing and map raw radio signals to fine-grained heartbeat waveforms. Also, a domain-transform neural network was introduced to transform radar-measured mechanical motion into electrocardiogram (ECG) signals [33], achieving accurate HRV monitoring.

Despite the promising results achieved by these methods, significant limitations remain. First, radio signals are highly sensitive to surrounding reflections. In particular, the continuous reflections across the body surface inherently introduce multiple interference sources, even when only the target is present. Most existing approaches rely on analyzing signals from single body reflection, which are highly susceptible to dynamic interference changes during monitoring. As a result, the signal can easily become blurred and distorted, leading to the loss of HRV details, as illustrated in Fig. 1(a). Second, heartbeat patterns vary significantly across different physiological conditions, resulting in a diverse distribution of HRV in real-world scenarios. This diversity makes it challenging for deep learning models to extract features effectively, especially under sparsely represented conditions, as illustrated in Fig. 1(c). These limitations ultimately restrict existing methods to controlled laboratory settings with limited subjects, render-

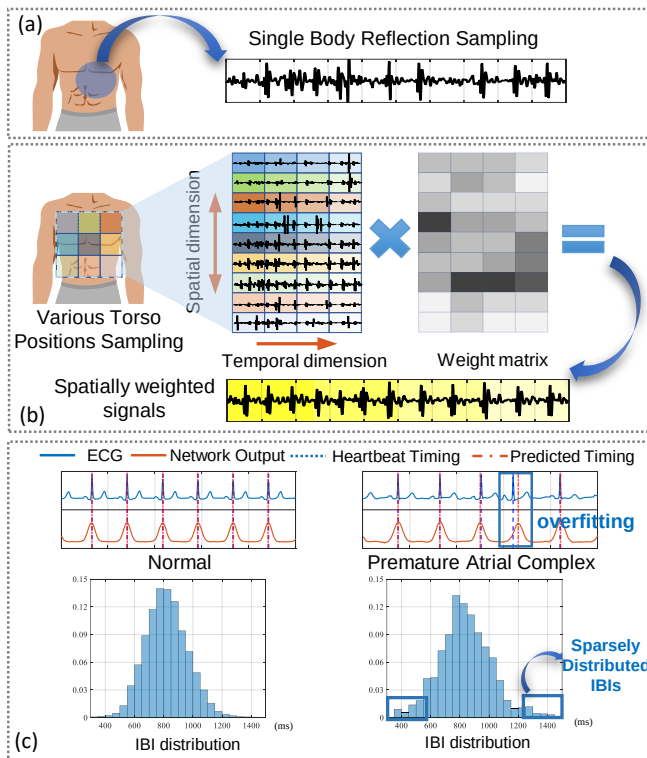


Fig. 1: Illustration of the limitations in current radar-based HRV monitoring and their relationship with physiological priors. (a) Analyzing signals from single body reflection are highly susceptible to interference. (b) Heartbeat motion is distributed across the torso surface, and spatially combining these signals with adaptive weights can enhance clarity. (c) Under normal conditions, where heartbeat intervals are more evenly distributed, the network learns effectively. However, sparsely distributed abnormal heartbeats, such as the occasional occurrence of premature atrial complexes, can lead to network overfitting.

ing them ineffective in real-world scenarios characterized by diverse physiological conditions. As a result, practical radar-based HRV monitoring remains an open challenge.

Fortunately, when digging into heartbeats, there are still physiological priors that can be leveraged to enhance radar measurements from both spatial and temporal dimensions, addressing the aforementioned challenges. Spatially, the inertial mechanical activity of the heart drives the entire torso surface to move with the same heartbeat rhythm while exhibiting spatial variations [39], [40]. Therefore, radar reflections from different torso locations carry cardiac motion information. Combining signals from these multiple torso locations can effectively suppress localized interference while reinforcing consistent cardiac patterns. To illustrate this intuition, we visualized radar measurements from various torso positions, as shown in Fig. 1(b). It is evident that while each measurement contains partial cardiac motion features, these features are occasionally obscured by interference. By combining these measurements using adaptive spatial weights (learned in our model), we can reconstruct a clear cardiac motion. Temporally, while HRV exhibits characteristic distributions under different physiological conditions [41], [42], cardiac motions still display self-similarity within individuals. This temporal consistency allows us to re-model HRV distributions across

varying conditions using existing signals, enabling efficient data-driven learning.

In this paper, we propose a physiological prior-inspired deep learning framework that integrates these priors into the radar signal domain for robust radar-based HRV monitoring. Specifically, we leverage the spatial redundancy prior that internal heartbeat motions drive movements across the entire torso surface and design a hybrid deep neural network adapted to the inherent noise, sparsity, and high-dimensional nature of radar signals. This network effectively models the inherent spatio-temporal relationship between radio reflections and heartbeat timing, dynamically leveraging full-body reflections while mitigating interference. Subsequently, we incorporate the prior of cardiac motion's temporal consistency along with the condition-specific HRV distribution characteristics of heartbeat intervals to develop a radar signal augmentation strategy. This strategy effectively creates various cardiac motion patterns through dynamic warping and re-models the data distribution during training, enabling more efficient data-driven learning. We built and evaluated our model on a large-scale, clinically oriented dataset comprising 7,150 outpatients collected from real hospital environments. Our method achieved a mean IBI error of 19.21 ms, an RMSSD error of 16.23 ms, an SDSD error of 16.70 ms, and a pNN50 error of 7.28%. on a validation set of 4,100 cases, reducing IBI errors by 34.46% compared to the baseline method. Additionally, we validated our method's HRV monitoring results in classifying five common arrhythmias, achieving performance comparable to gold-standard ECG-based HRV monitoring methods. Furthermore, ablation experiments demonstrated the effectiveness of each component of our system.

Our main contributions can be summarized as follows:

- 1) To the best of our knowledge, we are the first to reveal the significance of spatial redundancy and temporal consistency as physiological priors in radar-based HRV monitoring. Our work demonstrates how leveraging these priors can significantly enhance monitoring performance and improve signal robustness.
- 2) A deep learning framework that incorporates physiological priors into the radar signal domain for robust HRV monitoring is proposed. Within this framework, a dynamic neural network is designed to capture the spatio-temporal relationship between radar reflections and heartbeat dynamics. Additionally, a novel radar signal augmentation strategy is integrated to enable efficient data-driven learning by re-modeling data distributions during training.
- 3) We built and validated our method on a large-scale dataset comprising 7,150 outpatients collected in a real hospital setting. Experimental analysis demonstrated that our approach significantly outperforms baseline methods. Furthermore, cardiac condition classification based on HRV information extracted via our method closely aligns with the gold-standard ECG method, underscoring its robustness and clinical applicability.

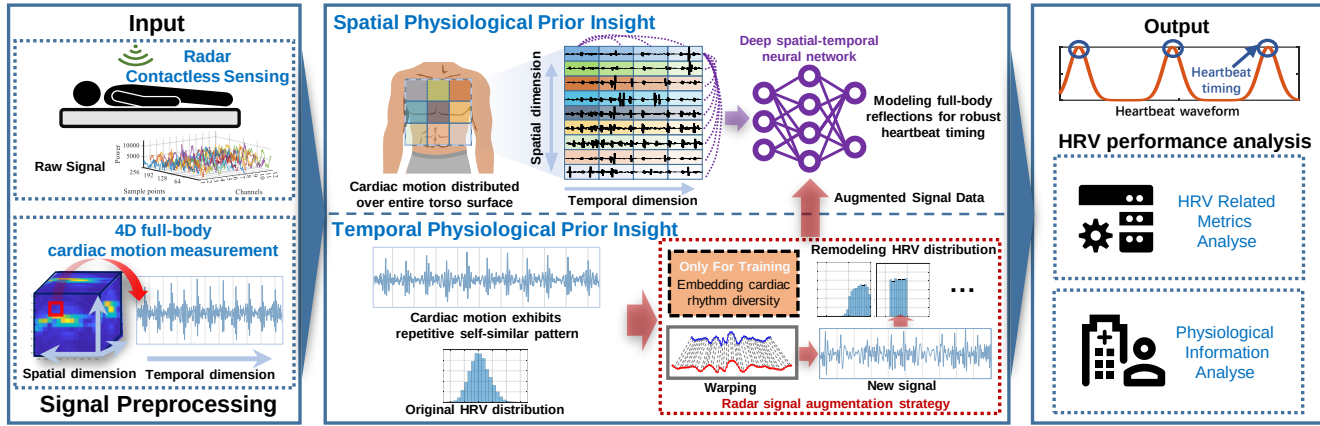


Fig. 2: System overview. The system begins with signal preprocessing, where raw radar signals are converted into 4D full-body cardiac motion measurements, capturing both spatial and temporal dimensions for detailed analysis. Leveraging spatial physiological priors, a deep spatio-temporal neural network is employed to model full-body reflections and accurately extract heartbeat timing information. To further enhance model robustness, temporal physiological priors are utilized to develop a radar signal augmentation strategy, which enriches the training data by embedding cardiac rhythm diversity and remodeling HRV distributions. Finally, the output waveforms are analyzed to extract HRV-related metrics and physiological insights, ensuring a comprehensive evaluation of system performance.

II. RELATED WORK

Radar-based HRV monitoring has gained increasing attention due to its ability to protect privacy while offering a comfortable user experience. Numerous efforts have been made to monitor heartbeat information using radar signals.

In those works, signal processing approaches focus on analyzing body surface movements caused by internal cardiac motion to extract HRV information. These methods process the modulated phase of reflected radio signals to achieve HRV estimation [24]–[27], [29]–[31]. For instance, Wang et al. utilized Variational Mode Decomposition (VMD) [43] to separate heartbeat motions from body surface signals [31]. Dong et al. filtered out respiratory signals and their harmonics to isolate accurate cardiac activity [26]. In addition, Dong et al. proposed a novel demodulation method to obtain more precise motion signals [25]. Also, Will et al. employed template matching to effectively extract body surface motion signals [27].

Recently, the advancement of deep learning has enabled researchers to make significant progress in directly translating radio signals into cardiac waveforms for HRV monitoring. [32]–[38]. For instance, Ha et al. recovered seismocardiograms (SCG) from radar signals [34]. Xu et al. reconstructed ECG-like waveforms from radar spectrogram [38]. Zhao et al. extracted heart sound envelopes from radar signals [35]. Additionally, Zhang et al. reconstructed accurate surface motion waveforms using deep learning and variational autoencoder networks [32], while Chen et al. introduced a system that converts mechanical motion signals from the body surface into surface electrophysiological signals [33]. However, Radio signals are highly sensitive to surrounding reflections. Existing methods primarily rely on analyzing signals from single torso reflection, making them susceptible to interference and resulting in low-quality input signals for neural networks, which in turn struggle to learn effective features. Additionally, heartbeat patterns vary significantly across physiological conditions, leading to diverse HRV distributions that pose challenges for deep learning models, particularly in sparsely represented

scenarios. These limitations confine current approaches to controlled laboratory settings with limited subject diversity (typically fewer than 50 subjects), hindering their deployment in real-world environments. Practical radar-based HRV monitoring, therefore, remains an open challenge.

In this work, we leverage the anti-interference capability of multi-point chest motion and the self-similarity of cardiac physiological motion as critical physiological priors. We design a spatial-temporal neural network for the complex radar signals near the chest to extract robust motion features. Furthermore, we enhance the diversity of the training data by reconstructing HRV distributions. Our system is validated on a large-scale real-world dataset, achieving accurate HRV monitoring. Also, through experiments, we demonstrate that the extracted HRV information provides arrhythmia-related physiological insights comparable to those from gold-standard ECG-based HRV.

III. SYSTEM DESIGN

In this section, we provide an overview of our system, as shown in Fig. 2. First, raw radar signals are preprocessed to spatially separate body reflection signals. Next, we design a neural network that leverages the spatial redundancy prior to model the relationship between full-body reflections and heartbeat timing. Finally, we enhance the diversity of cardiac rhythms in the training set through a radar signal augmentation strategy, utilizing the self-similarity of cardiac motion to remodel HRV distributions for efficient learning optimization.

A. Signal Preprocessing

To effectively utilize physiological prior knowledge, it is essential to extract cardiac motion across the entire body from raw signals in both spatial and temporal dimensions. We achieve this through several preprocessing steps, including spatial separation of reflection signals, torso reflection localization, and cardiac motion extraction.

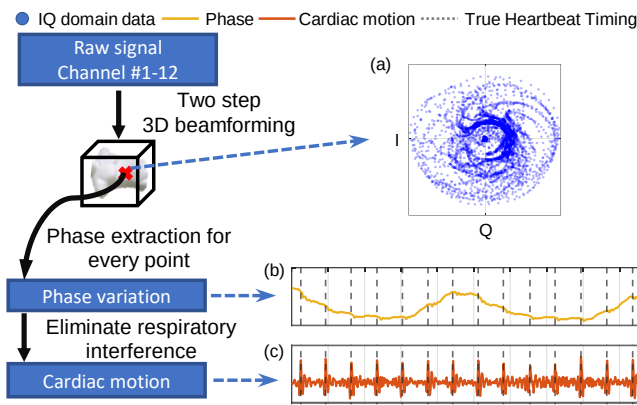


Fig. 3: Data flow for signal preprocessing. (a) Visualization of the IQ signal from a single voxel on the chest surface after beamforming. (b) Phase variation extracted from the IQ domain at the voxel. (c) Extracted cardiac motion after suppressing respiratory interference by the differentiator filter.

Spatial separation of reflection signals. We deploy a Frequency Modulated Continuous Wave (FMCW) millimeter-wave radar equipped with a 2D antenna array to comprehensively scan the sensing space. Using beamforming technology, the system separates reflections from different 3D voxels by mapping signal paths to their corresponding phase variations. Specifically, this is accomplished by combining the signals received on all channels (transmitter-receiver pairs) using the following equation [12]:

$$S_{(x,y,z,t)} = \sum_{n=1}^N \sum_{t_f=1}^T y_{n,t_f} e^{j2\pi(t_f-1)T_s k \frac{r(x,y,z,n)}{c}} e^{j2\pi \frac{r(x,y,z,n)}{\lambda}}, \quad (1)$$

where $S(x, y, z, t)$ represents the signal of voxel (x, y, z) at frame time t . N is the number of virtual channels, y_{n,t_f} is the received signal at t_f (ADC sampling points within the chirp), T_s is the ADC sampling period, k is the frequency slope, λ is the wavelength, $r(x, y, z, n)$ is the distance from voxel (x, y, z) to the antenna pair in channel n .

Torso reflection localization. After spatially separating the reflection signals, we need to locate the thoracic region within the signal space. We implement a two-step beamforming technique to accurately identify and extract reflections from the chest area. First, we apply coarse-grained grid beamforming to capture spatial signals over a wide field of view, followed by a Constant False Alarm Rate (CFAR) [44] detection method to coarsely localize body reflections. Next, we refine the focus on the detected point using fine-grained grid beamforming to achieve higher-resolution signals. The CFAR detector is then applied again to precisely locate the center of torso reflections. Finally, a predefined 3D bounding box is deployed to cover these signals, defining the thoracic reflection region for subsequent analysis.

Cardiac motion extraction. Radar signals can penetrate clothing and capture subtle body surface movements. The core principle of using radar sensors for detecting these movements lies in extracting phase variations from the received signals reflected off the target. The phase model is mathematically

defined as:

$$\phi(t) = 2\pi \frac{d(t)}{\lambda}, \quad (2)$$

where λ represents the wavelength of the radar signals, $d(t)$ signifies the distance between the radar sensor and the reflection object, and t denotes the sensing time.

Body surface movements include both heartbeat-related motion and respiratory motion, as shown in Fig. 3(b). To enhance heartbeat-related motion while suppressing respiratory interference, we apply temporal filtering for every torso reflection sampling. Specifically, we adopt the approach of extracting motion acceleration using a second-order differentiator in [13]:

$$s''_0 = \frac{(s_{-3} + s_3) + 2(s_{-2} + s_2) - (s_{-1} + s_1) - 4s_0}{16h^2}, \quad (3)$$

where s''_0 refers to the second derivative at a particular time sample, s_i is the value of the time series i samples away, h is the frame periodicity between consecutive samples. After applying time filtering, we are able to obtain the cardiac motion information detected by radar, as shown in Fig. 3(c).

Through the above processing steps, we transform the raw radar signals into a 4D representation of full-body cardiac motion in the radar signal domain. These processed measurements can be expressed in the form as $X_{in} \in R^{H \times W \times D \times L}$, where H , W and D represent the three spatial dimensions, L denote temporal length. This structured representation provides a foundation well-suited for integrating physiological priors in the subsequent deep learning framework design.

B. Deep Spatial-Temporal Neural Network

Our work is motivated by jointly analyzing body reflections to eliminate interference to acquire clear cardiac motion information. Heartbeat motion on the body surface is driven by internal heart activity and propagates across the torso in a non-rigid manner, with same cardiac rhythm while exhibit spatial variations [39], [40]. Effectively capturing this spatial redundancy requires an integrated modeling approach across the entire torso surface. However, radar signals, even after preprocessing, inherently contain dominant noise and interference, leaving the valuable cardiac motion information sparse. Additionally, HRV monitoring requires a high temporal sampling rate. With a sampling rate of 100 Hz over 10 seconds of signals, the temporal length reaches 1,000 samples. This high temporal resolution required for HRV monitoring, combined with the three-dimensional spatial resolution of body sampling, results in an extremely high-dimensional signal input size. Existing deep neural networks, primarily designed for tasks such as computer vision (CV) [45]–[47] and natural language processing (NLP) [48]–[50], struggle to address the unique characteristics of radar signals. To overcome these challenges, we propose a hybrid convolutional-transformer-based neural network architecture, as shown in Fig. 4, which achieves modeling for these 4D signal measurements by processing sequentially across temporal and spatial dimensions.

We first extract and compress temporal features independently from each spatial point. While Convolutional Neural Networks (CNNs) [45] have shown effectiveness in feature

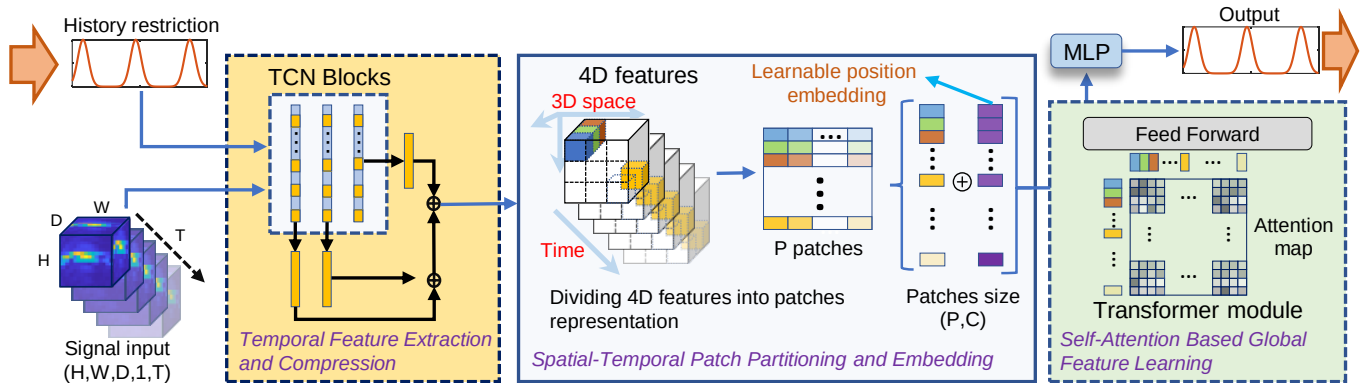


Fig. 4: Deep spatial-temporal neural network. The network leverages a conditional TCN to initially extract and compress spatiotemporal information from the input signals. The extracted 4D features, comprising spatial and temporal dimensions, are then divided into patches and enriched with learnable position embeddings to enhance representation capability. A transformer module further refines the global spatiotemporal cardiac features through self-attention mechanisms. Finally, an MLP maps the processed features to generate the cardiac waveform output.

extraction tasks, traditional CNNs struggle with the computational complexity posed by the large temporal input size of our signals. Temporal Convolutional Networks (TCNs) [46], on the other hand, provide significant advantages, including a wider receptive field and lower computational cost compared to traditional CNNs. Therefore, we adopt a TCN structure to extract and compress heartbeat-related temporal features from each spatial point efficiently first. In order to address the issue of explosive size in intermediate features caused by the large original input and increased channels, we sample only a portion of the features from each TCN layer. As shown in Fig. 4, this sampling is done with a size smaller than the dilation factor, ensuring that the sampled features still retain the receptive field of the entire input data. After passing through the TCN, the length of the temporal dimension T is compressed, while the channel dimension C is increased to represent the relevant features.

Due to the rhythmic nature of the heartbeat, which follows cyclic patterns influenced by ongoing physiological conditions, the current state of the heartbeat is inherently dependent on historical heartbeat patterns. To restrict the feature extraction process, we employ a Conditional Temporal Convolutional Network (TCN) [46], [51]. Within the TCN, Gated Activation Units (GAUs) are utilized to effectively capture and integrate historical features into the representation of the current heartbeat state, ensuring the temporal dependencies are preserved and leveraged during the extraction process.

However, directly using raw historical heartbeat data, as is commonly done in fields like audio processing, is unsuitable in this context due to the highly regular and repetitive nature of heartbeat signals. This regularity often results in overfitting, where the network memorizes static patterns rather than capturing the dynamic timing variations essential for accurate HRV monitoring. To address this issue, we introduce a history-blurring mechanism during training. Specifically, we apply a random masking and gaussian noise strategy to the historical heartbeat data. Random masking selectively hides parts of the historical information, forcing the network to infer the missing details based on the radar signals. Additionally, gaussian noise is added to further disrupt the regularity of the data, encouraging the model to learn robust features rather

than relying on the periodicity of the signal.

In the following, we need to jointly process features in 4D spatial-temporal dimensions. Inspired by Vision Transformers [47], [48], we convert multidimensional data into sequence-like patches. By grouping spatiotemporally neighboring features into the same patch via non-overlapping sum pooling, we partition the 4D space into P patches. Then, we employ a transformer module to capture the complex relationships between these patches. The input feature size of the transformer is $X_f \in R^{P \times C}$, where P represents the number of mixed spatio-temporal patches, while C represents the number of heartbeat-related features for each patch. The partitioned patches are enhanced with learnable position embeddings, which help capture spatial-temporal relationships and enhance their representation capability. These embedded patches are then processed by the transformer module, where self-attention mechanisms further refine the global spatiotemporal cardiac features. Finally, a two-layer Multi-Layer Perceptron (MLP) is applied to generate the heartbeat waveform output.

Our task focuses on accurate heartbeat timing rather than reconstructing full physiological waveforms. Inspired by Transrac [52], we generate the ground-truth heartbeat waveform by interpolating the time points of heartbeat events (R peaks in the ECG) with a Gaussian distribution, forming smooth Gaussian waveforms as the target signal (Fig. 5). A simple peak-finding process is then applied to the final output waveform to locate heartbeat timings.

C. Prior based radar signal augmentation strategy

The heart is regulated by the sinoatrial node, often referred to as the natural pacemaker of the heart. It generates electrical impulses that propagate through the heart's conduction system, causing rhythmic cardiac motions [53]–[56]. While these motions can vary significantly depending on factors such as body weight, cardiac health status, and heart size, the consistent electrical signaling ensures a repeating pattern of cardiac motion. This forms a self-similar temporal structure within the same individual, as illustrated in Fig. 6. This temporal self-similarity serves as a crucial physiological prior for analyzing cardiac status. For example, by segmenting the repeating QRS complexes in an ECG signal, individual cardiac cycles

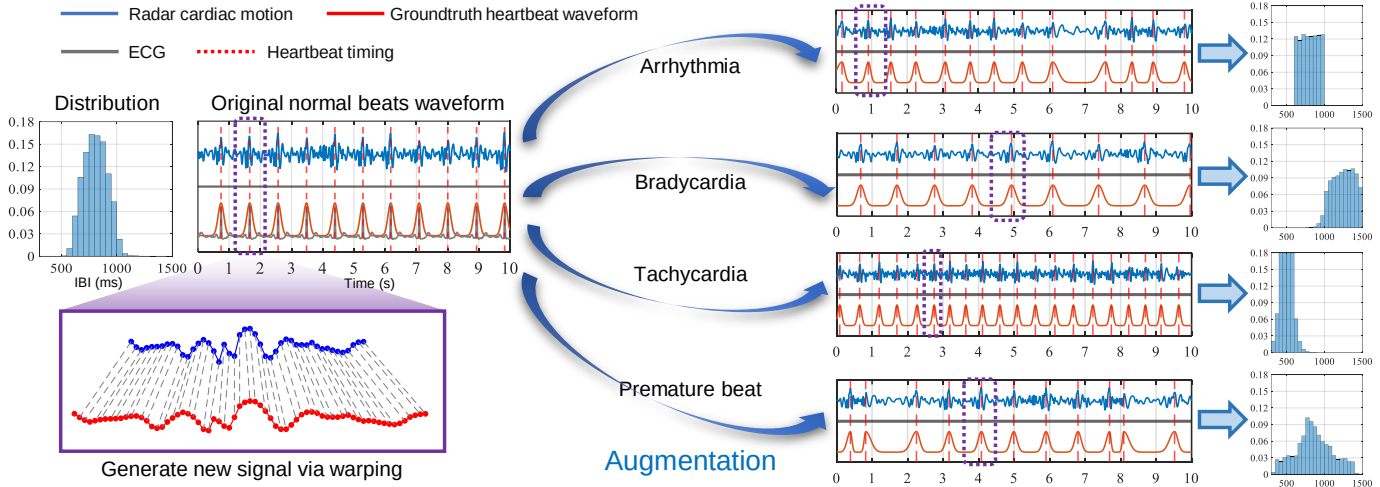


Fig. 5: Visualization of data augmentation strategies with warping illustration. The cardiac motion signal is obtained from a single spatial sampling point following the preprocessing step. The purple boxes highlight the changes in the same heartbeat before and after augmentation. The figure shows the original distribution of radar-detected cardiac motion, alongside the corresponding ground truth heartbeat waveform for original normal beats data. Different augmentation strategies, including arrhythmia, bradycardia, tachycardia, and premature beat, are applied to generate diverse heartbeat patterns. Each augmented signal is presented with its corresponding distribution to illustrate the changes introduced by the augmentation. Additionally, the bottom-left corner provides a detailed illustration of the warping-based augmentation process used to generate new cardiac signals.

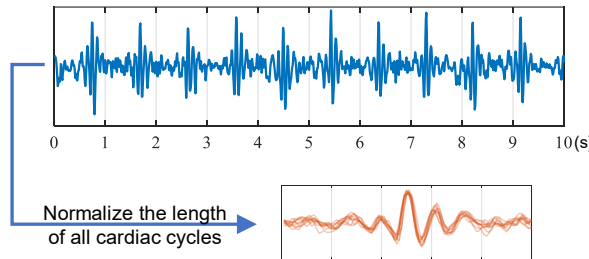


Fig. 6: Self-similarity of cardiac motion. By segmenting a period of cardiac motion from the same individual into individual cardiac cycles and normalizing their lengths, a consistent self-similar pattern in cardiac motion can be observed across cycles.

can be isolated and analyzed effectively. Taking this insight further, we leverage this self-similarity prior to build a signal augmentation strategy for HRV distribution re-modeling.

The distribution pattern of HRV is controlled by heart rhythms, which are significantly influenced by various physiological factors [41], [42]. From a clinical analysis perspective, these rhythms are commonly categorized into five major types based on heart rate and the regularity of heart rate patterns [41], [57]–[59] as follows:

Normal rhythm shows moderate and stable cycles (60–100 bpm), **tachycardia** involves shortened cycles (>100 bpm), and **bradycardia** presents prolonged cycles (<60 bpm). **Arrhythmia** is marked by irregular and unstable intervals, while **premature beats** typically display a short–long cycle pattern, where the early beat is followed by a compensatory pause.

Based on rhythm categories, we re-model the HRV distribution by warping cardiac cycle lengths to match characteristic patterns. For each radar–ECG pair, radar data are segmented into cardiac cycles using ECG R-peaks. Given $n + 1$ R-peaks, the signal is divided into $S = [S_0, S_1, \dots, S_{n+1}]$, where $S_1, \dots, S_n$ are n complete cycles with lengths $I = [I_1, I_2, \dots, I_n]$. These are resampled to new lengths $I' = [I'_1, I'_2, \dots, I'_n]$ according to target rhythm patterns. The warping is implemented by interpolation in the time domain:

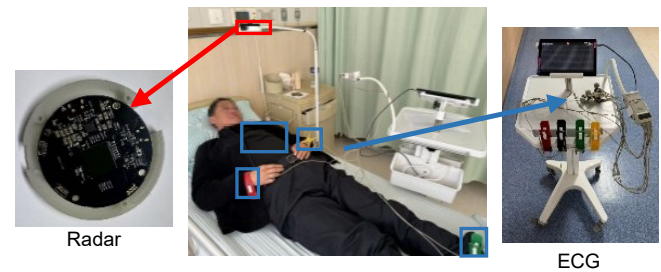


Fig. 7: Data collection setting. Radar and ECG were recorded simultaneously while the participant lay supine.

$$k = \frac{m \times (I_x - 1)}{I'_x - 1}, \quad (4)$$

$$S'_x[m] = x[[k]] + (k - [k]), (x[[k]] - x[[k]]), \quad (5)$$

where k is the interpolated index and $S'_x[m]$ the augmented value. The enhanced signal is $S' = [S_0, S'_1, \dots, S'_n, S_{n+1}]$, with updated IBI ground truth. All operations are time-domain only, preserving spatial structure.

Since normal rhythm dominates the dataset, we augment with four patterns—tachycardia, bradycardia, arrhythmia, and premature beats—using IBI ranges and pNN50, the percentage of successive IBI that differ by more than 50 milliseconds, to regulate stability. We visualize the distribution and waveform of data augmentation results in Fig. 5. The rules are:

- **Tachycardia:** $I'_i \in [40, 60]$, $\text{pNN50} < 20$
- **Bradycardia:** $I'_i \in [100, 150]$, $\text{pNN50} < 20$
- **Arrhythmia:** $I'_i \in [60, 100]$, $\text{pNN50} > 20$
- **Premature beats:** Only a part of paired IBI in each record is resampled. $I'_i = I_i - l$, $I'_{i+1} = I_{i+1} + l$, ensuring $I_i - l \geq 40$.

This warping is applied only during training as a physiologically inspired augmentation to broaden IBI distribution and enhance robustness, and is not used at inference.

IV. EXPERIMENTS AND RESULTS

A. Experimental Setup

We build and validate our deep learning model on a large-scale dataset collected in a real-world clinical environment. This dataset includes synchronized radar and ECG recordings from 7,150 outpatient visitors during their routine 30-second ECG screenings in a hospital. During data collection, participants lay on a bed with the radar positioned approximately 60 cm above their chest, as shown in Fig. 7. Heartbeat timing ground truth was derived using NeuroKit2 [60], an open-source toolkit for ECG processing, to detect R-peaks in synchronized ECG signals. Additionally, All recordings were annotated by experts to provide reference cardiac conditions based on ECG analysis. As the dataset was collected from outpatient visitors, it offers substantial diversity in cardiac conditions. Given the extensive range of conditions, we structurally categorized them into three main groups: Normal, Common Arrhythmias, and Others. The dataset demographics are presented in Table I. This diverse and clinically representative dataset serves as a robust foundation for evaluating our model's performance, particularly in terms of HRV monitoring and its medical applicability in real-world usage scenarios.

We adopted subject-wise stratified 3-fold cross-validation. In each run, one fold served as the test set. The remaining two folds were further partitioned, with 70% of the data allocated to training and 30% to validation. Stratification preserved the distribution of diagnostic labels across folds, and subject-level partitioning prevented any subject overlap between the training/validation or test sets. Accordingly, we report performance aggregated across all folds on the entire dataset.

B. Implementation Detail

During experiments, the 12-lead ECG data is collected, filtered, and output by the MedEx ECG analysis system, ultimately obtaining 12-lead ECG data at a sampling rate of 500Hz. We utilized the TI1843 AOP radar [61] to collect corresponding radar signals. Three transmitting antennas and four receiving antennas were employed to simulate a 12-channel 2D virtual antenna array. The configuration details of chirp and frames are the same as mmECG [33]. In such a configuration, the radar achieved a frame rate of 100Hz and a bandwidth of 3.32GHz.

Our spatial beamforming was implemented based on the relative coordinates of the radar. The redundant space near the chest was initially set at a coarse granularity with a size of $(x = 1.5m) \times (y = 1.5m) \times (z = 1.5m)$, resulting in 343 voxels $(7 \times 7 \times 7)$. This was then refined to a smaller space with a size of $(x = 0.5m) \times (y = 0.5m) \times (z = 0.5m)$, resulting in 729 voxels $(9 \times 9 \times 9)$. The x-axis is parallel to the body width, the y-axis is parallel to the body height, and the z-axis is perpendicular to the chest surface. Considering the size of the human chest, our input space is designed to be redundant, facilitating the extraction of more comprehensive spatiotemporal features by the network.

In our network implementation, we utilized a three-layer conditional TCN with a dilation factor of 8 to extract heartbeat

TABLE I: Cardiac condition abbreviations and dataset distribution

Cardiac Condition	Abbreviation	Subjects
Common Arrhythmias	CA	1513
Atrial flutter	AFL	22
Sinus arrhythmia	SA	101
Atrial fibrillation	AF	136
Ventricular premature beats	PVC	166
Premature atrial contractions	PAC	187
Sinus tachycardia	ST	266
Sinus bradycardia	SB	722
Non-common Arrhythmias	NCA	5637
Localized chest low voltage	LCLV	4
Complete left bundle branch block	CLBBB	25
Counterclockwise rotation	CCR	48
Abnormal Q wave	Qab	50
Clockwise rotation	CR	69
Limb lead low voltage	LLLV	72
Left ventricular high voltage	LVHV	101
Complete right bundle branch block	CRBBB	156
T-wave abnormality	TAb	979
ST-segment abnormality	STAb	1568
Sinus rhythm	SR	2542
Normal ECG	NECG	3313
Total		7150

features from spatial points, achieving a receptive field of size 512 at the final layer. After each TCN layer, features were sampled at an interval of 5, compressing the time dimension to 128, while the channel dimension was set to 128, transforming the radar input of size $729 \times 1 \times 640$ into preliminary features of size $729 \times 128 \times 128$. These features were further divided into 864 spatiotemporal patches using a sum pooling technique with a spatial kernel size of $3 \times 3 \times 3$ and a temporal kernel size of 4, both with corresponding steps. A six-layer transformer encoder was then applied to extract heartbeat-related information from the patches, with each block having 4 attention heads ($Q/K/V = 64$) and a feed-forward dimension of 128. The spatial dimensions of the transformer output were summed, yielding global features of size 32×128 . Finally, a two-layer MLP mapped the global features to a 100-frame heartbeat waveform, with historical heartbeat waveforms used as constraints. During training, followed by mmECG [33], we build historical segments from ECG-derived heartbeat waveforms, adding Gaussian noise scaled to twice the signal standard deviation and randomly masking the first 0–100 % of the sequence to simulate noisy or incomplete history and improve generalization. During inference, the first segment is zero-initialized, and subsequent segments are updated by rolling and concatenating previous predictions, enabling our autoregressive model to maintain temporal continuity.

For data augmentation, due to the self-similarity of heartbeat motion waveforms within each person, we apply IBI resampling data augmentation to each recording in the training dataset. Specifically, 1/3 of the recordings are augmented using the arrhythmia pattern, 1/2 using the premature beat pattern, 1/12 for the tachycardia pattern, and 1/12 for the bradycardia

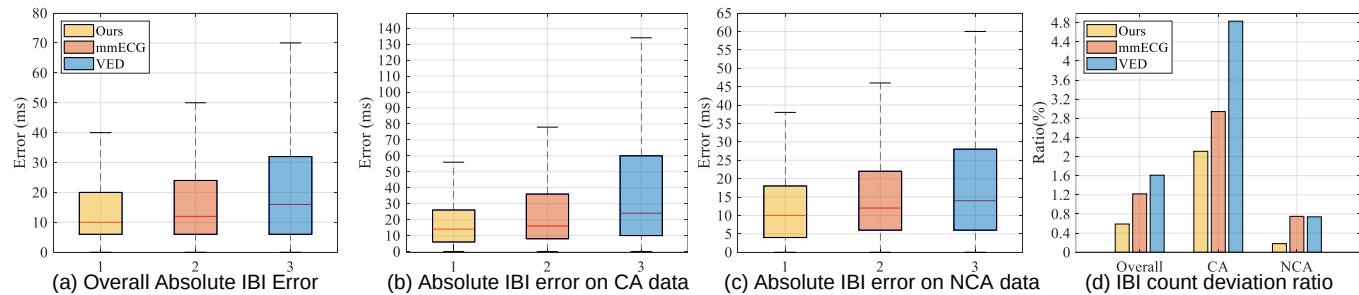


Fig. 8: Performance across different cardiac conditions: The IBI error is evaluated for the overall dataset, as well as for common arrhythmias (CA) and non-common arrhythmias (NCA).

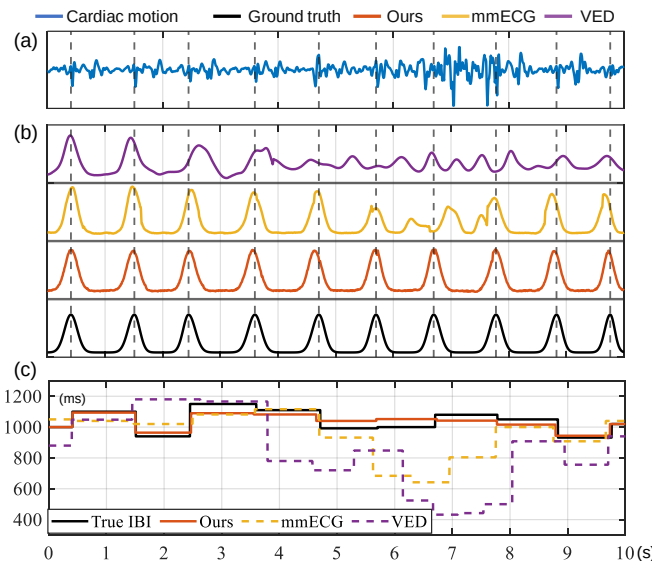


Fig. 9: Impact of wrong heartbeats output on IBI accuracy. (a) Visualization of the cardiac motion from a single point within the network input. (b) Comparison of the output of different networks with the ground truth, with rows corresponding to VED, mmECG, our method, and the ground truth, respectively. (c) Analysis of the impact of the number of incorrect IBIs along the sampling dimension on IBI performance.

pattern. Premature beats are given the highest probability due to their lower occurrence rate in the data, while arrhythmia is prioritized to help the network learn more about radar signals' heart-related information and rhythm patterns. Considering training costs, the total number of augmented recordings is kept equal to the original training recordings in this section.

C. Baselines

We employed VED [32] and mmECG [33] as our baseline methods. VED applies deep learning techniques to recover heart-related waveforms, while mmECG utilizes deep learning to transform mechanical motion detected by radar into ECG waveform. Both methods are designed to extract detailed HRV information effectively.

D. Metrics

HRV features are calculated based on IBI sequences, making IBI errors a crucial indicator of system performance. To ensure fair comparisons and eliminate ambiguity in IBI error calculations caused by differences in heartbeat counts between the generated and ground truth data, we employ the metric of **real-time absolute IBI error**. Unlike traditional methods,

this metric calculates IBIs along the time sampling dimension rather than relying on heartbeat sequences, as illustrated in Fig. 9(c). HRV features are generally classified into three categories: time-domain, frequency-domain, and non-linear features, each capturing different aspects of heart rate variability. In this study, we focus on three of the most commonly used HRV indicators to evaluate accuracy: **RMSSD**: Root mean square of successive differences, which reflects short-term variability in heart rate. **SDSD**: Standard deviation of successive differences, indicating overall variability in heart rate regulation. **pNN50**: Percentage of successive IBIs differing by more than 50 ms, closely associated with parasympathetic nervous system activity.

E. Results

1) Overall performance: We first evaluated the system's overall performance in terms of IBI recovery. Common IBI performance analysis is typically conducted on the whole dataset. As shown in Fig. 8(a), the median IBI errors for VED, mmECG, and our method are 16 ms, 12 ms, and 10 ms, respectively, with mean errors of 38.34 ms, 29.31 ms, and 19.21 ms. However, since normal heartbeat data constitutes the majority of the dataset, overall performance is primarily influenced by the accuracy on normal rhythms. This makes it difficult to evaluate the performance of these methods on data with more diverse HRV distributions. Therefore, we divided the dataset into two parts for further analysis: common arrhythmias (CA) recordings, which include seven common types of arrhythmias, and non-common arrhythmia (NCA) recordings, which consist of the remaining data with relatively normal heart rates.

The performance of the three methods on CA data is shown in Fig. 8(b), with median errors of 24 ms, 16 ms, and 14 ms, and mean errors of 77.23 ms, 47.38 ms, and 34.20 ms, respectively. On NCA data, as shown in Fig. 8(c), the median errors are 14 ms, 12 ms, and 10 ms, with mean errors of 27.55 ms, 24.30 ms, and 15.06 ms, respectively. In CA data, our method achieves a mean IBI error reduction of 13.18 ms compared to the best baseline, which is significantly more substantial than the reductions on the overall dataset (10.10 ms) and NCA data (9.24 ms). This highlights the superior performance of our method on HRV data with more diverse abnormal heart rate distributions.

The poorer performance of baseline methods compared to our method may be attributed to their limitations in extracting cardiac motion features or overfitting, which often result in false positives or missed heartbeats. This frequently leads to

TABLE II: Cardiac condition specific and Sex specific performance.

Errors	IBI(ms)↓			RMSSD(ms)↓			SDSD(ms)↓			PNN50(%)↓			CDR(%)↓			Subjects
	VED	mmECG	Ours	VED	mmECG	Ours	VED	mmECG	Ours	VED	mmECG	Ours	VED	mmECG	Ours	
Conditions																
NECG	27.57	23.86	14.29	26.43	36.97	11.82	27.04	37.63	12.14	10.98	11.15	6.68	0.67	0.70	0.12	3313
SR	33.48	28.28	18.14	36.00	42.73	18.05	36.83	43.53	18.57	11.05	11.02	6.85	1.03	0.94	0.32	2542
STAb	51.52	36.60	25.94	44.82	43.24	20.68	45.95	44.18	21.31	13.36	12.46	8.02	2.92	1.91	1.30	1568
TAb	45.92	34.96	24.21	40.22	43.93	19.41	41.21	44.81	19.96	13.01	12.24	7.93	2.42	1.77	1.16	979
SB	52.64	32.80	22.94	49.01	34.85	20.72	50.44	35.74	21.41	16.91	14.12	11.19	2.69	1.76	0.82	722
ST	49.41	30.16	19.36	57.67	47.01	15.51	58.54	47.72	15.79	13.44	11.15	4.37	3.80	2.05	1.14	266
PAC	89.01	56.89	34.30	92.70	62.63	39.32	94.99	64.37	40.43	16.70	13.14	7.40	3.56	2.57	1.26	187
PVC	118.60	79.07	55.40	137.92	100.36	67.79	141.09	102.87	69.52	21.10	12.66	8.09	5.83	3.55	2.57	166
CRBBB	54.18	34.64	26.77	49.94	42.26	21.48	51.08	43.20	22.09	13.45	12.87	7.83	3.13	1.44	1.28	156
AF	216.02	120.44	103.42	98.29	68.53	44.90	101.10	70.53	46.21	25.09	15.81	8.96	16.14	9.63	9.41	136
LVHV	69.03	49.19	43.66	64.01	55.68	33.68	65.70	57.08	34.63	16.57	14.29	10.35	4.97	3.66	3.50	101
SA	68.00	30.96	19.41	52.81	33.57	19.34	54.18	34.38	19.88	18.78	13.85	10.54	3.04	0.55	0.05	101
LLLV	42.16	27.77	23.79	38.13	27.43	18.18	39.12	27.88	18.75	12.84	10.12	7.44	2.79	2.56	2.10	72
CR	27.41	24.68	15.18	27.99	34.09	15.01	28.64	34.72	15.42	9.65	10.46	8.13	0.65	0.76	0.00	69
QAb	46.43	27.89	21.94	47.07	33.97	15.98	48.16	34.57	16.47	13.12	12.18	9.52	1.59	0.77	0.57	50
CCR	28.95	22.37	15.66	28.82	36.87	13.41	29.47	37.44	13.72	9.46	11.14	7.83	0.63	0.56	0.28	48
CLBBB	52.17	36.26	21.99	49.36	39.11	23.61	50.52	40.45	24.22	11.85	10.24	8.68	1.68	0.58	0.00	25
AFL	234.42	155.77	132.22	143.49	81.56	57.93	148.28	84.72	60.03	27.85	25.91	14.87	19.75	14.22	13.27	22
LCLV	59.57	24.05	13.32	76.36	44.55	5.15	78.89	45.70	5.35	19.31	22.45	4.46	1.92	0.00	0.00	4
CA	77.23	47.38	34.20	67.90	48.98	28.34	69.60	50.17	29.15	17.61	13.66	9.16	4.83	2.94	2.11	1513
NCA	27.55	24.30	15.06	26.84	37.47	12.86	27.46	38.12	13.23	10.71	11.03	6.76	0.74	0.75	0.18	5637
Male	40.99	30.15	20.27	38.16	39.83	17.02	39.13	40.64	17.52	12.52	11.58	6.93	1.76	1.28	0.67	3246
Female	36.16	28.63	18.33	33.80	40.09	15.58	34.57	40.84	16.02	11.96	11.62	7.58	1.49	1.16	0.52	3904
Overall	38.34	29.31	19.21	35.77	39.97	16.23	36.63	40.75	16.70	12.21	11.60	7.28	1.61	1.22	0.59	7150

significantly anomalous IBIs, impacting the accurate recovery of IBIs and causing usability challenges, as shown in Fig. 9. To illustrate this, we utilize the IBI count deviation ratio (CDR) to reflect the system's stability in recovering the number of IBIs. The formula for CDR is as follows:

$$CDR = \frac{|N_p - N_t|}{N_t} \times 100\%, \quad (6)$$

where N_p represents the predicted number of IBIs and N_t represents the true number of IBIs from the ECG data.

As shown in Fig. 8(d), we compared the CDR of different methods across various data types. Our method outperforms the baselines in all cases. On the overall dataset, our method achieves a CDR of 0.59%, compared to the best baseline's 1.22%. On CA data, our method achieves a CDR of 2.11%, an improvement of 0.84% over the best baseline, and on NCA data, our method achieves a CDR of 0.18%, an improvement of at least 0.56%. These results highlight the superior stability of our system in terms of IBI count recovery, reflecting higher recovery accuracy and reliability.

2) Cardiac condition specific performance: To delve deeper into the system's performance across diverse physiological conditions in real-world scenarios, we conducted a detailed analysis of the performance of various methods on different cardiac conditions. Our results, as illustrated in Table II, unequivocally demonstrate that our method performs better across various error metrics regardless of different cardiac condition.

For the NECG (Normal ECG, conditions with only sinus rhythm) data, our method achieves a mean absolute IBI error of 14.29 ms. And for the SR (sinus rhythm) data, our method achieves a mean absolute IBI error of 18.14 ms. On STAb (ST-segment abnormality) and TAb (T-wave abnormality) data, the mean IBI errors are 25.94 ms and 24.21 ms, respectively. For SB (sinus bradycardia), our method achieves 22.94 ms, while

for ST (sinus tachycardia), the mean error is 19.36 ms. On PAC (premature atrial contractions) data, the mean error is 34.30 ms, and for PVC (premature ventricular contractions), the error is 55.40 ms. On CRBBB (complete right bundle branch block), our method achieves a mean error of 26.77 ms, while for AF (atrial fibrillation) data, the mean error is 103.42 ms. On LVHV (left ventricular high voltage), the error is 43.66 ms, and for SA (sinus arrhythmia), it is 19.41 ms.

Further, for LLLV (limb lead low voltage), our method achieves an impressive mean IBI error of 23.79 ms, while for CR (clockwise rotation), the error is 15.18 ms. On QAb (abnormal Q wave) data, the error is 21.94 ms, and for CCR (counterclockwise rotation), it is 15.66 ms. For CLBBB (complete left bundle branch block), our method achieves a mean error of 21.99 ms, while for AFL (atrial flutter), the error is 132.22 ms. Finally, on LCLV (localized chest low voltage), our method achieves a mean IBI error of 13.32 ms.

Our method demonstrates superior performance across all cardiac conditions, consistently outperforming the two baseline methods in both IBI recovery and HRV indicators such as RMSSD, SDSD, and pNN50 (one-sided paired Wilcoxon tests, $p < 0.001$ for both baselines). This highlights the system's robustness and its ability to provide precise and reliable HRV feature extraction. These results validate the effectiveness of integrating physiological priors, which significantly enhance the accuracy of IBI recovery and HRV metrics.

3) Sex specific performance: To examine whether each method performs stably across sex, we conducted a sex-stratified analysis; the results are shown in Table II. Across methods, the performance patterns remain consistent, and our approach maintains its advantage in both cohorts.

Specifically, our method shows comparable performance across sexes, with mean IBI errors around 20 ms in both cohorts. The mean error was 18.33 ms in females and 20.27

TABLE III: HRV features used in arrhythmia classification experiment.

Feature extraction methods	Features
Time-domain	MeanNN,SDNN,RMSSD,SDSD,CVNN, CVSD,MedianNN,MedNN,MCVNN,IQRNN
Nonlinear	pNN50,pNN20,TINN,HTI

ms in males, with females exhibiting slightly lower error. Also, other HRV metrics (e.g., RMSSD, SDSD, pNN50) show similar trends. We examined this discrepancy and found a plausible contributor to be case mix: approximately 25% of the male cohort presented common arrhythmias, compared with about 18% in females. These findings indicate that the proposed system is robust across sex and suitable for practical deployment.

F. Evaluating System Performance Through HRV-based Arrhythmia Classification

Existing radar-based HRV monitoring methods primarily focus on achieving accuracy in HRV metrics. However, these numerical results alone provide limited insight into their practical value in real-world clinical scenarios. Since HRV serves as a reflection of various physiological states associated with cardiac conditions, we conducted an experiment to evaluate whether our HRV results can effectively differentiate between arrhythmia types, thereby assessing their potential to capture diagnostic insights. Specifically, inspired by Asl et al. [62], we employed an SVM model to evaluate whether the HRV features extracted from our system can effectively differentiate between cardiac conditions, compared to the gold-standard ECG-based HRV method. The selected time-domain and nonlinear HRV indicators are summarized in Table III.

We use five categories of cardiac conditions for classification: sinus tachycardia (ST), sinus bradycardia (SB), atrial fibrillation or atrial flutter (AF/AFL), and premature atrial or ventricular contractions (PAC/PVC). Additionally, we define the Normal class as recordings with normal rhythm, excluding cases with common arrhythmias.

We trained multiple SVM classifiers to validate the system's ability to distinguish different cardiac conditions individually. The training process employed a 3-fold cross-validation strategy, and the final performance was determined by averaging results across the three test sets. The classification threshold was set at the point where the F1 score, a key performance indicator, achieved its optimal value. Evaluation metrics, including sensitivity, specificity, accuracy, F1 score, and Area Under the Curve (AUC), were used to assess classification performance. Additionally, Receiver Operating Characteristic (ROC) curves were plotted for each category to provide a detailed comparison of classification performance based on HRV data.

The classification performance is presented in Table IV. As summarized in the Table, our radar-derived HRV achieves accuracies of 0.975 (SB), 0.987 (ST), 0.982 (AF/AFL), 0.942 (PAC/PVC), and 0.901 (Normal). Across all categories, these results outperform both radar baselines (mmECG and VED) and are close to the upper-bound provided by ECG-derived HRV.

TABLE IV: Classification performance of obtained HRV.

Metrics($\pm$ std)	Classes	SB	ST	AF/AFL	PA/VC	Normal
Accuracy $\uparrow$	ECG	0.975 $\pm$ 0.003	0.986 $\pm$ 0.001	0.986 $\pm$ 0.005	0.961 $\pm$ 0.008	0.929 $\pm$ 0.011
	Ours	0.975 $\pm$ 0.003	0.987 $\pm$ 0.001	0.982 $\pm$ 0.005	0.942 $\pm$ 0.004	0.901 $\pm$ 0.007
	mmECG	0.971 $\pm$ 0.002	0.986 $\pm$ 0.001	0.980 $\pm$ 0.003	0.913 $\pm$ 0.027	0.876 $\pm$ 0.009
	VED	0.972 $\pm$ 0.002	0.984 $\pm$ 0.004	0.964 $\pm$ 0.008	0.884 $\pm$ 0.014	0.862 $\pm$ 0.005
	ECG	0.982 $\pm$ 0.005	0.989 $\pm$ 0.002	0.989 $\pm$ 0.006	0.974 $\pm$ 0.007	0.832 $\pm$ 0.032
Specificity $\uparrow$	Ours	0.982 $\pm$ 0.004	0.992 $\pm$ 0.001	0.984 $\pm$ 0.006	0.959 $\pm$ 0.003	0.747 $\pm$ 0.001
	mmECG	0.979 $\pm$ 0.002	0.990 $\pm$ 0.002	0.980 $\pm$ 0.005	0.935 $\pm$ 0.033	0.631 $\pm$ 0.039
	VED	0.981 $\pm$ 0.004	0.988 $\pm$ 0.005	0.971 $\pm$ 0.008	0.904 $\pm$ 0.015	0.631 $\pm$ 0.056
	ECG	0.914 $\pm$ 0.014	0.899 $\pm$ 0.050	0.852 $\pm$ 0.056	0.657 $\pm$ 0.026	0.961 $\pm$ 0.022
	Ours	0.921 $\pm$ 0.006	0.838 $\pm$ 0.016	0.886 $\pm$ 0.017	0.564 $\pm$ 0.085	0.952 $\pm$ 0.010
Sensitivity $\uparrow$	mmECG	0.902 $\pm$ 0.004	0.836 $\pm$ 0.026	0.762 $\pm$ 0.079	0.406 $\pm$ 0.106	0.948 $\pm$ 0.002
	VED	0.912 $\pm$ 0.028	0.823 $\pm$ 0.005	0.659 $\pm$ 0.051	0.419 $\pm$ 0.064	0.945 $\pm$ 0.014
	ECG	0.882 $\pm$ 0.010	0.821 $\pm$ 0.015	0.747 $\pm$ 0.058	0.591 $\pm$ 0.057	0.953 $\pm$ 0.008
	Ours	0.884 $\pm$ 0.013	0.821 $\pm$ 0.003	0.687 $\pm$ 0.060	0.451 $\pm$ 0.046	0.935 $\pm$ 0.005
	mmECG	0.865 $\pm$ 0.005	0.819 $\pm$ 0.010	0.649 $\pm$ 0.023	0.287 $\pm$ 0.011	0.921 $\pm$ 0.005
F1 score $\uparrow$	VED	0.880 $\pm$ 0.008	0.807 $\pm$ 0.040	0.447 $\pm$ 0.044	0.234 $\pm$ 0.032	0.911 $\pm$ 0.002
	ECG	0.990 $\pm$ 0.001	0.984 $\pm$ 0.007	0.977 $\pm$ 0.011	0.855 $\pm$ 0.020	0.957 $\pm$ 0.005
	Ours	0.989 $\pm$ 0.002	0.983 $\pm$ 0.006	0.974 $\pm$ 0.011	0.836 $\pm$ 0.011	0.923 $\pm$ 0.003
	mmECG	0.987 $\pm$ 0.001	0.983 $\pm$ 0.012	0.972 $\pm$ 0.010	0.774 $\pm$ 0.014	0.886 $\pm$ 0.005
	VED	0.988 $\pm$ 0.002	0.981 $\pm$ 0.014	0.949 $\pm$ 0.027	0.733 $\pm$ 0.010	0.863 $\pm$ 0.005

Fig. 10 further compares ROC curves of radar-derived HRV with ECG-derived HRV (upper bound) across multiple cardiac conditions. Our approach achieves strong discrimination with AUCs of 0.989 for SB, 0.983 for ST, 0.974 for AF/AFL, 0.836 for PAC/PVC, and 0.923 for Normal, also clearly outperforming existing radar-based baselines and approaching the ECG upper bound (0.990/0.984/0.977/0.855/0.957).

Additionally, Table IV reports more precise metrics together with their variance estimates; across metrics, all variances of our system are less than 0.09 and most are less than 0.05. Taken together, the system demonstrates classification performance that aligns closely with the ECG method across a range of cardiac conditions and shows good generalization, highlighting its reliability and potential for real-world applications.

Besides, to investigate the issue of false negative and false positive classifications, we analyzed the t-SNE visualization and examined the performance of both false positive and false negative data. The analysis further explained two possible causes for the errors. First, as HRV is a broad-spectrum physiological indicator, its features associated with different cardiac conditions often overlap, leading to inherent ambiguity in HRV-based classification. This overlap limits the discriminatory power of HRV alone in precisely distinguishing specific cardiac states. Second, a significant portion of the misclassified data may either be in a transitional state between positive and negative samples or contain errors in manual labeling. Taking sinus bradycardia as an example, in Fig. 11, the false negative data shows an average IBI of 1028ms, which closely aligns with the standard for sinus bradycardia. Similarly, the false positive data in Fig. 11 should actually be categorized under sinus bradycardia. This misclassification arises because the labeling is performed manually, and ambiguity during this process can cause instability in the network's predictions under such states. Despite these factors, our system still maintains performance levels closely aligned with ECG-based methods, demonstrating its robustness and reliability in extracting meaningful physiological insights.

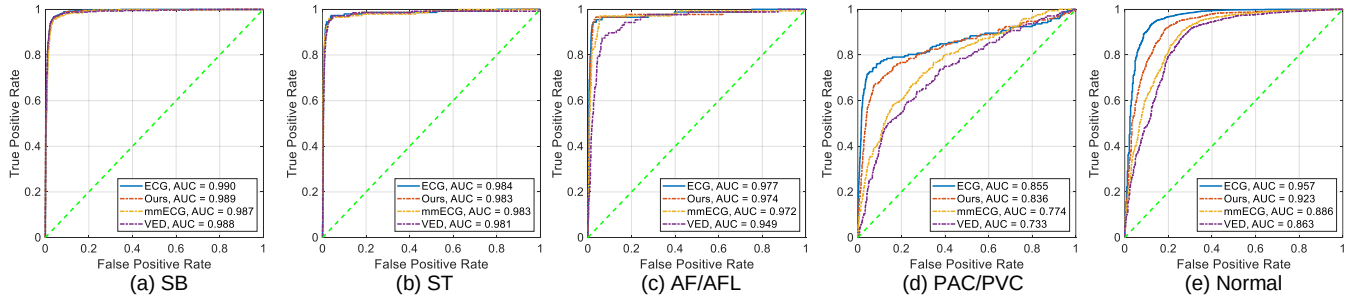


Fig. 10: Receiver operating characteristic curves comparing radar-derived HRV with ECG-derived HRV (upper-bound) for classification of cardiac conditions. (a) SB: sinus bradycardia, (b) ST: sinus tachycardia, (c) AF/AFL: atrial fibrillation and atrial flutter, (d) PAC/PVC: premature atrial and ventricular contractions, and (e) Normal: normal rhythm (excluding cases with common arrhythmias).

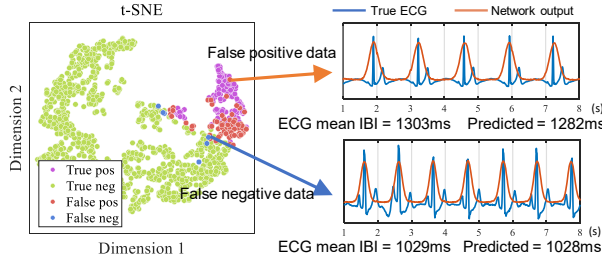


Fig. 11: The t-SNE visualization of sinus bradycardia recordings. Sinus bradycardia refers to the recording whose ECG mean IBI is greater than 1000ms.

TABLE V: Frequency domain HRV Analysis

Errors	meanIBI(ms)↓	VLF(ms ²)↓	LF(ms ²)↓	HF(ms ²)↓
VED	35.82	876.75	1147.94	1408.59
mmECG	22.24	89.79	217.26	318.77
our	13.07	54.15	35.77	53.60

G. Frequency domain HRV Analysis

Frequency-domain HRV provides rich and complementary information to time-domain indices, as it reflects autonomic modulation and respiratory coupling in cardiac rhythms. To evaluate our method in this domain, we conducted an additional experiment focusing on frequency-domain HRV features. The outpatient recordings used in our primary analysis are relatively short (about 30 seconds) and thus unsuitable for spectral analysis. Therefore, we collected longer recordings specifically for this purpose. In this experiment, 10 subjects underwent 20-minute radar-ECG acquisitions, and evaluation was performed using the model trained on the outpatient dataset. Standard spectral indices were analyzed, including Very Low Frequency (VLF, 0.003–0.04 Hz), Low Frequency (LF, 0.04–0.15 Hz), and High Frequency (HF, 0.15–0.40 Hz).

As summarized in Table V, our approach yields the lowest errors across all bands: mean IBI error of 13.07 ms (vs. 22.24 ms for mmECG, 35.82 ms for VED); VLF error of 54.15 ms² (vs. 89.79 ms² for mmECG, 876.75 ms² for VED); LF error of 35.77 ms² (vs. 217.26 ms² for mmECG, 1147.94 ms² for VED); and HF error of 53.60 ms² (vs. 318.77 ms² for mmECG, 1408.59 ms² for VED). These results indicate that our lower beat-to-beat timing error translates into smaller spectral errors in VLF/LF/HF, and that the proposed method remains superior to both baselines in frequency domain HRV as well.

H. Nearby Human Interference Analysis

Nearby human presence is common in practical deployments. We conducted an additional experiment using the

TABLE VI: Ablation settings and results.

Ablation patterns			Errors					
Spatial signal	Spatial attention	Resample modes	IBI(ms)↓			CDR(%)↓		
			CA	NCA	Overall	CA	NCA	Overall
-	-	-	75.51	25.70	36.43	6.90	1.13	2.33
✓	-	-	68.56	22.67	32.55	4.34	0.30	1.14
✓	✓	-	43.34	16.04	21.91	2.55	0.21	0.69
✓	✓	Random	91.78	38.65	50.05	9.35	2.27	3.74
✓	✓	Partten	33.08	15.10	18.97	1.79	0.17	0.50

model trained on the outpatient dataset to evaluate the robustness of our system to nearby human interference. Specifically, we collected data from 10 subjects, each with three 5-min sessions under different conditions: no human interference, interference at 0.5 m, and interference at 0.2 m. We then evaluated the changes in IBI error across these settings.

As shown in Fig. 13, our method maintains stable performance under most conditions: no interference—median IBI error 10 ms (mean 13.87 ms); 0.5 m—median 12 ms (mean 15.11 ms); 0.2 m—median 26 ms (mean 81.59 ms). These results indicate that very close interference (0.2 m) can severely affect IBI estimation, whereas at 0.5 m—the more common scenario in practical use—the performance remains essentially unaffected. Notably, the 0.5 m setting already satisfies the requirements of most real-world applications, underscoring the robustness of our system to nearby human interference.

I. Ablation Study

In this section, we conduct a series of ablation experiments to demonstrate the significance of leveraging physiological priors and evaluate the effectiveness of designs, including the spatio-temporal network architecture and signal augmentation strategy.

We first perform ablation experiments on different input signal sizes, spatial attention and remodeling strategies. All ablations here use a fixed data fold for controlled, like-for-like comparisons. The detailed setting and results are presented in Table VI. In the first row, we replace the multi-position sampling signals with a single-reflection signal using the VED method and discard both the spatial attention module and the HRV remodeling module; this serves as a baseline against the second row, where multi-position sampling signals are employed without the attention or remodeling modules, demonstrating the advantage of leveraging spatial-redundancy priors to comprehensively model reflections from different torso positions. In the third row, we enable the spatial attention

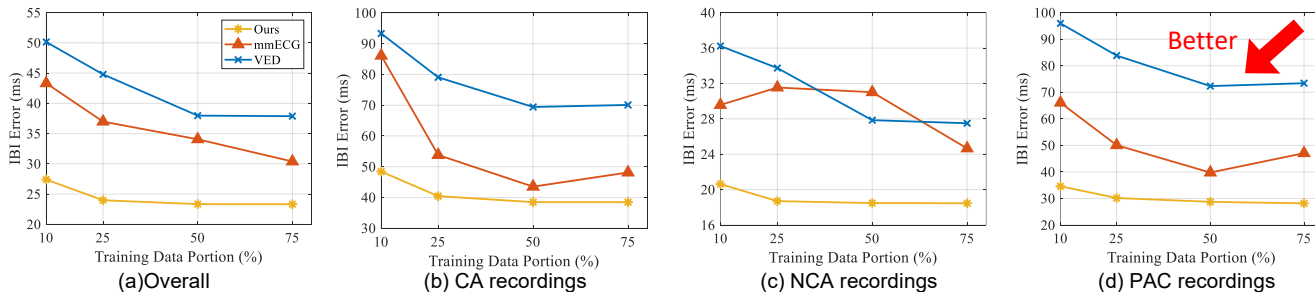


Fig. 12: Performance under different training data portions. Each subfigure shows the inter-beat interval (IBI) estimation error of our method (yellow), mmECG (red), and VED (blue) when varying the percentage of training data used. (a) Overall; (b) common arrhythmias (CA); (c) non-common arrhythmias (NCA); (d) premature atrial contractions (PAC). Lower values indicate better performance.

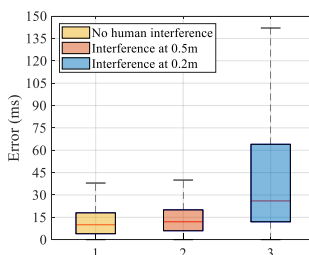


Fig. 13: Nearby human interference analysis

module but do not perform resampling, and the comparison with the second row supports the necessity of our spatial-attention design. In the fourth row, we randomly remodel the IBI sequence within a valid range (40–150 ms), which disrupts the network’s ability to fit normal heartbeat patterns and causes a noticeable performance drop. The final row emphasizes remodeling based on physiological prior knowledge in the IBI-sequence dimension; the significant improvement validates our remodeling strategy. Collectively, these results confirm the importance of physiological priors and the effectiveness of our design in enhancing the robustness and accuracy of radar-based HRV monitoring.

Furthermore, we evaluate the performance of our system across varying training set sizes to assess the effectiveness of the signal augmentation strategy in enhancing model robustness and generalization. Specifically, we reserved approximately 25% of the recordings for testing and used different proportions of the remaining dataset for training, including 10%, 25%, 50%, and 75%. The signal augmentation strategy was applied consistently across all training subsets. This setup allows us to analyze the robustness of our system under varying data availability conditions. For comparison, we benchmark our model against baseline models to highlight the improvements achieved through our augmentation approach.

The experimental results, as shown in Fig. 12, illustrate the overall performance, as well as the performance on CA and NCA recordings, with a focus on a representative cardiac condition from the CA recordings. As the training dataset size increased, all methods exhibited improvements in overall performance. However, when training data became abundant, the performance gains of baseline networks were primarily driven by overfitting on NCA recordings, while their performance on CA recordings plateaued or even declined. This trend is particularly evident in the performance of PAC recordings. In contrast, our method demonstrated consistent stability and superior performance across all training set sizes

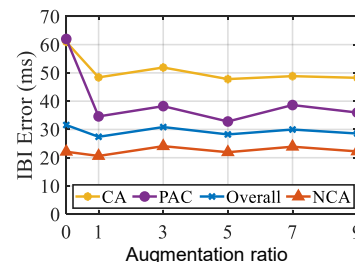


Fig. 14: Performance under different augmentation ratios.

when compared to the baseline models. Notably, our approach excelled in scenarios with limited training data, achieving better fitting performance without compromising robustness. These findings highlight our system’s ability to effectively utilize signal augmentation strategies, ensuring both stability and enhanced generalization across diverse data conditions.

To further validate the robustness and effectiveness of the proposed data augmentation module, we conducted experiments to analyze the impact of varying augmentation ratios K , defined as the multiple of augmented data relative to the original dataset. Specifically, the analysis was performed under extremely limited training data scenarios, using only 10% of the total recordings as the original training set.

As shown in Fig. 14, the results demonstrate that all augmentation ratios significantly enhance performance on CA recordings, underscoring the module’s ability to improve learning under data scarcity. However, higher augmentation ratios slightly affect the model’s fit NCA recordings, leading to a minor decrease in overall performance. These findings highlight the balance between augmentation ratio and model generalization, demonstrating that the proposed augmentation module achieves substantial performance improvements even with lower augmentation ratios. This experiment reinforces the effectiveness and adaptability of the augmentation module, particularly in addressing challenges posed by limited training data, further supporting the design choices introduced in our framework.

J. Model Complexity Analysis and Real-Time Feasibility

We profiled the deployed model using a 1-second output window and a batch size of one on an NVIDIA RTX 4090 GPU. The model contains 2.37 million trainable parameters and has a model size of 9.05 MB. Each forward pass requires 35.48 GFLOPs and completes in 14.48 milliseconds per 1-second sample, roughly 69 times faster than real time. Because the pipeline needs only one forward pass per second, these

measurements demonstrate real-time operation with ample headroom for preprocessing and peak detection, confirming the system's suitability for real-time applications.

V. DISCUSSION

The present study introduces a physiological prior-inspired deep learning framework for radar-based HRV monitoring, with significant technical and clinical impact. Specifically, the study explains the spatio-temporal prior information essential for radar HRV monitoring and proposes a novel methods for effectively applying such priors.

The system offers a convenient and effective solution for contactless HRV monitoring, with the potential to provide deeper insights into disease progression, support early detection, personalize treatment plans, and enhance preventive care through continuous, long-term monitoring. Despite its potential, the clinical impact and transformative potential of this novel technology for long-term HRV monitoring, particularly its unique advantages in enabling ultra-long-term, operation-free, and contactless everyday monitoring, still require further exploration and validation. We aim to make this a key focus of our future research, evaluating its performance in real-world ultra-long-term settings, developing adaptive real-time analysis techniques, and collaborating with clinicians to assess and optimize its integration into clinical workflows.

VI. CONCLUSIONS

In this study, we introduced a novel system for extracting precise human heart rate variability by leveraging physiological prior knowledge. By extracting the spatio-temporal relationship between full-body radio reflections and heartbeat timing through deep neural networks, and remodeling training data distributions via a radar signal enhancement strategy, we developed a robust and stable HRV detection system. Validated on large-scale, diverse datasets from real-world clinical scenarios, our system demonstrates superior performance compared to baseline methods. Moreover, it effectively captures HRV-related physiological insights, achieving results comparable to ECG-based methods. These findings underscore the practical value of our approach for non-contact HRV monitoring in real-world healthcare applications.

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